

A two-site fluctuation theorem for the boundary-driven resonant NLS diffusion

Existence of the NESS, finite-time path relations, and interior SCGF symmetry

Working proof note

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Abstract

For the two-site projection-free Cartesian resonant NLS diffusion, both sites are directly thermostatted and the diffusion matrix is uniformly elliptic. We exploit this feature to close analytic assumptions that remain open for a long chain. The quartic energy yields an exponential Foster–Lyapunov function, hence a unique nonequilibrium steady state with a smooth strictly positive density and exponential energy moments. Time reversal then gives the finite-time forward–reverse fluctuation relation for total entropy. For every tilt $k \in (0, 1)$, a second exponential Lyapunov estimate verifies the compactness and minorization hypotheses for the medium-entropy Feynman–Kac kernel. It has a simple principal growth exponent $\psi_2(k)$, and the exact adjoint identity implies

$$\psi_2(k) = \psi_2(1 - k), \quad 0 < k < 1.$$

The theorem is continuous-time and model-specific. It does not assert a same-forward-ensemble detailed relation at finite time, endpoint regularity at $k = 0, 1$, or a thermodynamic-limit theorem.

1 Model and statement

Write $c_j = x_j + iy_j$, $I_j = |c_j|^2$, $M = I_1 + I_2$, and $z = (x_1, y_1, x_2, y_2) \in \mathbb{R}^4$. The physical energy is

$$E(z) = \frac{1}{2}M^2 - \frac{1}{4}(I_1^2 + I_2^2) + \operatorname{Re}(c_2^2 \overline{c_1^2}). \quad (1)$$

Let $F_j = \nabla_j E$, let J be block-diagonal with the standard two-dimensional symplectic block, and let $T_1, T_2, \gamma > 0$. The two-site diffusion is

$$dz_t = J \nabla E(z_t) dt - \gamma \nabla E(z_t) dt + \sqrt{2\gamma} D_T^{1/2} dW_t, \quad D_T = \operatorname{diag}(T_1, T_1, T_2, T_2). \quad (2)$$

Its generator is

$$\mathcal{L} = (J \nabla E) \cdot \nabla - \gamma \sum_{r=1}^2 F_r \cdot \nabla_r + \gamma \sum_{r=1}^2 T_r \Delta_r. \quad (3)$$

Complex conjugation defines the time reversal

$$\Theta(x_1, y_1, x_2, y_2) = (x_1, -y_1, x_2, -y_2), \quad (\mathcal{T}f)(z) = f(\Theta z).$$

For heat positive into the system, set

$$dQ_r = F_r \circ \left[-\gamma F_r dt + \sqrt{2\gamma T_r} dW_{r,t} \right], \quad \Sigma_t^m = -\frac{Q_1(t)}{T_1} - \frac{Q_2(t)}{T_2}. \quad (4)$$

For $k \in \mathbb{R}$, define the positive kernel

$$P_{t,k}f(z) = \mathbb{E}_z \left[e^{-k\Sigma_t^m} f(Z_t) \right]. \quad (5)$$

Theorem 1.1 (Two-site fluctuation theorem). *The diffusion (2) has a unique invariant probability measure μ with a smooth strictly positive density ρ . If $T_{\max} = \max(T_1, T_2)$, then*

$$\int e^{aE(z)} \mu(dz) < \infty, \quad 0 < a < T_{\max}^{-1}. \quad (6)$$

For every finite time, the total entropy

$$\Sigma_t^{\text{tot}} = \Sigma_t^m + \log \rho(Z_0) - \log \rho(Z_t) \quad (7)$$

obeys the forward–reverse Crooks relation and $\mathbb{E}_\mu[e^{-\Sigma_t^{\text{tot}}}] = 1$.

For every $k \in (0, 1)$, the kernel (5) admits a simple positive principal growth exponent and

$$\psi_2(k) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E}_\mu e^{-k\Sigma_t^m} \quad (8)$$

exists and is finite. It satisfies

$$\boxed{\psi_2(k) = \psi_2(1 - k), \quad 0 < k < 1.} \quad (9)$$

The remaining sections prove the theorem in the order: recurrence, finite-time path reversal, principal Feynman–Kac growth, and adjoint symmetry.

2 Quartic geometry and an exponential Lyapunov function

Lemma 2.1 (Coercivity and gradient coercivity). *The energy is a homogeneous polynomial of degree four and*

$$E \geq \frac{1}{4}(I_1^2 + I_2^2) \geq \frac{1}{8}M^2. \quad (10)$$

Moreover,

$$|\nabla E|^2 \geq \frac{16E^2}{M} \geq \frac{1}{4}M^3 \quad (M > 0). \quad (11)$$

At either site, $\Delta_r E = 4M$.

Proof. Writing $c_j = \sqrt{I_j} e^{i\varphi_j}$ gives

$$E = \frac{1}{4}(I_1^2 + I_2^2) + I_1 I_2 \{1 + \cos[2(\varphi_2 - \varphi_1)]\},$$

which proves (10). Every term in (1) is quartic, so Euler’s identity gives $z \cdot \nabla E = 4E$. Since $|z|^2 = M$, Cauchy–Schwarz and (10) imply

$$|\nabla E|^2 \geq \frac{16E^2}{M} \geq \frac{16(M^2/8)^2}{M} = \frac{1}{4}M^3.$$

The boundary-Laplacian identity follows by direct differentiation; for $n = 2$ both sites are boundary sites. $\square$

Lemma 2.2 (Exponential drift). *For $W_a = e^{aE}$ and $0 < a < T_{\max}^{-1}$,*

$$\frac{\mathcal{L}W_a}{W_a} = -\gamma a \sum_{r=1}^2 (1 - aT_r) |F_r|^2 + 4\gamma a(T_1 + T_2)M, \quad (12)$$

and therefore

$$\frac{\mathcal{L}W_a}{W_a} \longrightarrow -\infty \quad \text{as } |z| \rightarrow \infty. \quad (13)$$

In particular, there are $\lambda, b > 0$ and a compact set C such that

$$\mathcal{L}W_a \leq -\lambda W_a + b\mathbf{1}_C. \quad (14)$$

Proof. The Hamiltonian vector field annihilates every function of E . Applying the two bath generators to e^{aE} gives (12). Put $\delta = 1 - aT_{\max} > 0$. By lemma 2.1,

$$\frac{\mathcal{L}W_a}{W_a} \leq -\frac{\gamma a \delta}{4} M^3 + 4\gamma a(T_1 + T_2)M,$$

which proves both claims. $\square$

Proposition 2.3 (Unique nonequilibrium steady state). *The claims about μ and ρ in the first sentence of theorem 1.1, including (6), hold. In fact the process is geometrically ergodic in a W_a -weighted norm for every $0 < a < T_{\max}^{-1}$.*

Proof. The diffusion covariance $2\gamma D_T$ is constant and strictly positive definite. Smoothness and uniform ellipticity give the strong Feller property and a continuous strictly positive transition density. Hence every compact set is petite, the process is topologically irreducible and its skeleton is aperiodic. The Foster–Lyapunov estimate (14) then gives positive Harris recurrence, a unique invariant probability measure, and weighted geometric ergodicity by the continuous-time Foster–Lyapunov theorem [1]. Integrating the drift inequality under invariant truncations gives $\mu(W_a) < \infty$. Finally, elliptic regularity applied to $\mathcal{L}^*\rho = 0$ gives $\rho \in C^\infty$, and positivity of the transition density gives $\rho(z) > 0$ everywhere. $\square$

Remark 2.4. The uniformly elliptic step is special to $n = 2$: both sites are endpoints. For $n \geq 3$ the covariance is degenerate, so this argument cannot simply be copied to a long chain.

3 Finite-time total-entropy relation

Lemma 3.1 (Finite stationary entropy production). *The invariant law has all polynomial energy moments and finite Fisher information in the bath metric:*

$$\sum_{r=1}^2 \gamma T_r \int |\nabla_r \log \rho|^2 \rho \, dz < \infty. \quad (15)$$

Proof. Polynomial moments follow from (6). Multiply the stationary Fokker–Planck equation by a compactly truncated $\log \rho$ and integrate by parts. The Hamiltonian vector field has zero divergence, while $\text{div}(-\gamma \nabla E) = -\gamma \Delta E$. Sending the cutoff to infinity using the exponential moment yields

$$\sum_r \gamma T_r \int |\nabla_r \log \rho|^2 \rho \, dz = \gamma \int \Delta E \rho \, dz < \infty.$$

Here $\Delta E = 8M$, so the final integral is finite. $\square$

Let $\mathfrak{R}$ act on a path by $(\mathfrak{R}\omega)_s = \Theta\omega_{t-s}$. Denote the stationary forward path law by $\mathbb{P}$ and its conjugate reverse by $\mathbb{P}^R = \mathbb{P} \circ \mathfrak{R}^{-1}$, with the reversed drift written in the standard stationary time-reversal form.

Proposition 3.2 (Crooks and integral fluctuation relations). *On every finite interval, $\mathbb{P}$ and $\mathbb{P}^R$ are mutually absolutely continuous and*

$$\log \frac{d\mathbb{P}}{d\mathbb{P}^R} = \Sigma_t^m + \log \rho(Z_0) - \log \rho(Z_t) = \Sigma_t^{\text{tot}}. \quad (16)$$

Consequently,

$$\mathbb{E}_\mu e^{-\Sigma_t^{\text{tot}}} = 1, \quad (17)$$

and, as measures on $\mathbb{R}$,

$$\mathbb{P}(\Sigma_t^{\text{tot}} \in ds) = e^s \mathbb{P}^R(\Sigma_t^{\text{tot}} \in -ds). \quad (18)$$

Proof. Uniform ellipticity, smooth positive stationary density, and lemma 3.1 place the stationary diffusion in the finite-entropy class for which the reversed path is again a diffusion and the forward/reverse Radon–Nikodym derivative is well defined [2, 3]. The Hamiltonian part is divergence-free and odd under Θ , hence contributes no irreversible path action. The two additive Langevin baths give the local-detailed-balance increments $-dQ_r/T_r$ in the Stratonovich convention. The initial and terminal stationary densities supply the remaining endpoint ratio, proving (16). Equation (17) is normalization of the Radon–Nikodym derivative; applying $\mathfrak{R}$ gives (18). $\square$

Remark 3.3 (What is not claimed at finite time). Equation (18) compares forward and conjugate-reverse ensembles. Unless the nonequilibrium stationary path law is itself invariant under $\mathfrak{R}$, it does not reduce to $P_F(s)/P_F(-s) = e^s$ within one forward ensemble.

4 Principal Feynman–Kac growth

The Itô product formula for (5) gives

$$\mathcal{L}_k = (J\nabla E) \cdot \nabla + \sum_{r=1}^2 \left[\gamma T_r \Delta_r + \gamma(2k-1)F_r \cdot \nabla_r + \gamma\beta_r(k^2 - k)|F_r|^2 + \gamma k \Delta_r E \right]. \quad (19)$$

Lemma 4.1 (Multiplicative Lyapunov estimate). *Fix $k \in (0, 1)$ and choose*

$$0 < a < \frac{1-k}{T_{\max}}. \quad (20)$$

Then

$$\begin{aligned} \frac{\mathcal{L}_k W_a}{W_a} &= \gamma \sum_{r=1}^2 \left[(T_r a + k) \left(a - \frac{1-k}{T_r} \right) |F_r|^2 \right. \\ &\quad \left. + 4(aT_r + k)M \right] \longrightarrow -\infty \quad \text{as } |z| \rightarrow \infty. \end{aligned} \quad (21)$$

If $0 < a' < a$, then $W_{a'}/W_a \rightarrow 0$ and $\mathcal{L}_k W_{a'}/W_{a'}$ is bounded above.

Proof. Apply (19) to e^{aE} and factor the coefficient of $|F_r|^2$:

$$T_r a^2 + (2k-1)a + \frac{k^2 - k}{T_r} = (T_r a + k) \left(a - \frac{1-k}{T_r} \right).$$

The first factor is positive and the second is uniformly negative under (20). The negative multiple of $|\nabla E|^2$ dominates the $O(M)$ term by (11). The same argument applies to a' , while $e^{-(a-a')E} \rightarrow 0$ by coercivity. $\square$

Proposition 4.2 (Principal entropy-tilted exponent). *For every $k \in (0, 1)$, $P_{t,k}$ has a continuous strictly positive kernel and a simple positive principal eigenvalue $e^{t\psi_2(k)}$ on a suitable W_a -weighted space. For every probability ν with $\nu(W_a) < \infty$,*

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \nu(P_{t,k} \mathbf{1}) = \psi_2(k). \quad (22)$$

In particular (8) holds.

Proof. Localize (5) on energy sublevel sets. Applying Itô's formula to $e^{-k\Sigma_t^m} W_{a'}(Z_t)$ and using the upper bound in lemma 4.1 gives a nonnegative local supermartingale after a deterministic exponential compensation. Fatou's lemma removes the stopping times and proves that the positive kernel is finite and bounded on the stated weighted space. Thus (5) is globally defined by the original nonexplosive diffusion; no potentially explosive change-of-drift process is needed. Local parabolic regularity for the uniformly elliptic principal part of (19), followed by the strong maximum principle, gives a continuous strictly positive density. Hence every compact set satisfies a minorization condition.

The two conclusions of lemma 4.1 are precisely the continuous multiplicative Lyapunov and majorization conditions for a positive Feynman–Kac semigroup: W_a supplies compactness at infinity and $W_{a'}$ controls the unbounded weight. The Lyapunov–minorization theorem for Feynman–Kac semigroups [4] therefore gives a unique normalized eigenmeasure, a simple strictly positive eigenfunction, exponential convergence of the normalized semigroup, and (22). The invariant law μ is admissible because proposition 2.3 gives $\mu(W_a) < \infty$. $\square$

Lemma 4.3 (L^2 compact realization). *Fix $k \in (0, 1)$. The closure of $\mathcal{L}_k$ on $L^2(\mathbb{R}^4, dz)$ has compact resolvent after a sufficiently large scalar shift. Its spectral bound is the same simple real eigenvalue $\psi_2(k)$ obtained in proposition 4.2. Moreover, the Hilbert-space adjoint is the closure of the formal adjoint on $C_c^\infty(\mathbb{R}^4)$.*

Proof. For real $f \in C_c^\infty$ (and by polarization for complex f), integration by parts gives

$$\begin{aligned} \operatorname{Re}\langle f, -\mathcal{L}_k f \rangle &= \gamma \sum_{r=1}^2 T_r \|\nabla_r f\|_2^2 + \gamma k(1-k) \sum_{r=1}^2 \beta_r \int |F_r|^2 |f|^2 dz \\ &\quad - \frac{\gamma}{2} \sum_{r=1}^2 \int (\Delta_r E) |f|^2 dz. \end{aligned} \quad (23)$$

The Hamiltonian part is skew because it is divergence-free. The coefficient $1/2$ in the last term follows from $-\frac{1}{2}(2k-1) + k = \frac{1}{2}$ and is independent of k . Using $\sum_r \Delta_r E = 8M$ and

$$\sum_r \beta_r |F_r|^2 \geq T_{\max}^{-1} |\nabla E|^2 \geq (4T_{\max})^{-1} M^3,$$

the M^3 term dominates the negative $O(M)$ term. Thus, for a sufficiently large C_k , there is $c_k > 0$ such that

$$\operatorname{Re}\langle f, (C_k - \mathcal{L}_k) f \rangle \geq c_k \left(\|f\|_{H^1}^2 + \int M^3 |f|^2 dz \right). \quad (24)$$

Let

$$\mathcal{V} = H^1(\mathbb{R}^4) \cap L^2(\mathbb{R}^4, M^3 dz).$$

The polynomial bounds $|\nabla E| \leq CM^{3/2}$ and $|\Delta E| \leq C(1 + M)$ show that the shifted nonsymmetric form $\langle f, (C_k - \mathcal{L}_k)g \rangle$ is continuous on $\mathcal{V} \times \mathcal{V}$; for example,

$$\left| \int \bar{f} (J\nabla E) \cdot \nabla g \, dz \right| \leq C \|M^{3/2} f\|_2 \|\nabla g\|_2.$$

Estimate (24) makes that form coercive after increasing C_k if necessary. The closed-sectorial-form construction therefore gives the maximal L^2 realization and identifies its adjoint with the operator constructed from the adjoint form. Since $\mathcal{V} \hookrightarrow L^2$ is compact (Rellich on a ball and the M^3 term for the exterior tail), the resolvent is compact. The form semigroup is the positive parabolic kernel of proposition 4.2 by uniqueness of weak solutions. Strong positivity makes its leading eigenvalue simple, and the normalized-semigroup convergence in proposition 4.2 identifies that spectral bound with $\psi_2(k)$. Finally, integration by parts on the form core C_c^∞ identifies the Hilbert adjoint with the closure of the formal adjoint. This is the standard compact-embedding consequence of the representation theorem for closed sectorial forms [6]. $\square$

5 Gallavotti–Cohen symmetry

Lemma 5.1 (Semigroup adjoint relation). *For $k \in (0, 1)$, on compactly supported smooth functions,*

$$\mathcal{L}_k^* = \mathcal{T} \mathcal{L}_{1-k} \mathcal{T}. \quad (25)$$

The corresponding positive kernels satisfy

$$p_{t,k}(x, y) = p_{t,1-k}(\Theta y, \Theta x). \quad (26)$$

Proof. The Hamiltonian vector field is divergence-free and odd under Θ . For one bath, $(F_r \cdot \nabla_r)^* = -F_r \cdot \nabla_r - \Delta_r E$. Taking the adjoint changes the drift coefficient $\gamma(2k - 1)$ into $\gamma(2(1 - k) - 1)$, changes the coefficient of $\Delta_r E$ from γk to $\gamma(1 - k)$, and leaves $k^2 - k = (1 - k)^2 - (1 - k)$ unchanged. This proves (25). Uniqueness of the positive parabolic kernel in the Lyapunov class from lemma 4.1 then gives (26); this is the diffusion realization of the Lebowitz–Spohn adjoint mechanism [5]. $\square$

Proof of theorem 1.1. The NESS and finite-time assertions are propositions 2.3 and 3.2. Fix $k \in (0, 1)$. By lemma 4.3, both complementary tilted generators have compact L^2 realizations and their spectral bounds are respectively $\psi_2(k)$ and $\psi_2(1 - k)$. The involution $\mathcal{T}$ is unitary on $L^2(dz)$. Therefore (25), extended to the closed domains by lemma 4.3, shows that the two operators are related by unitary conjugation and Hilbert adjunction. Their spectra, and in particular their simple real spectral bounds, coincide. Hence $\psi_2(k) = \psi_2(1 - k)$, which proves (9). $\square$

6 Claim boundary and relation to the simulations

The result proves the continuous-time interior-tilt symmetry at $n = 2$. The numerical pairs (0.3, 0.7) and (0.4, 0.6) lie strictly inside the proved interval. The following are deliberately not folded into the theorem:

1. extending the SCGF statement to the endpoint moments $k = 0, 1$ without a separate endpoint-domain argument;
2. a same-forward finite-time detailed relation;

3. the degenerate-noise NESS and spectral theory for $n \geq 3$;
4. uniformity in n or a nontrivial $n \rightarrow \infty$ limit;
5. any statement about the time-discrete numerical integrator without a convergence estimate.

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